Document Object Model (DOM)

- The Document Object Model (DOM) represents an HTML page as a document that JavaScript can access and modify.

- JavaScript can use the DOM to read HTML content.
- JavaScript can change the content of HTML elements.
- JavaScript can create dynamic changes in a web page.
- The document object is used to work with the HTML document.

DOM Methods and Properties

The following methods and properties are important for basic DOM programming.

Methods

1. document.write()

The document.write() method writes content directly into the HTML document.

```
document.write("Hello Students");
```

Output: Hello Students

2. document.writeln()

The document.writeln() method writes text to the document and adds a newline character after the text.

```
document.writeln("Hello");
document.writeln("Students");
```

The method is similar to document.write(), but it writes a newline after the supplied text.

3. document.getElementById()

The getElementById() method finds an HTML element using its id attribute.

```
<p id="message">Hello</p>

<script>
   let element = document.getElementById("message");
   console.log(element);
</script>
```

The method returns the element whose id matches the supplied value.

Changing an Element Using getElementById()

```
<p id="message">Old Message</p>

<script>
   let element = document.getElementById("message");
   element.innerHTML = "New Message";
</script>
```

The content of the paragraph changes from Old Message to New Message.

4. document.getElementsByClassName()

The getElementsByClassName() method finds HTML elements that have a specified class name.

```
<p class="item">One</p>
<p class="item">Two</p>
<script>
   let elements = document.getElementsByClassName("item");
   console.log(elements);
</script>
```

The method returns to a collection of elements that have the specified class.

5. document.getElementsByTagName()

The getElementsByTagName() method finds HTML elements using their tag name, such as p, h1, div, or li.

```
<p>First Paragraph</p>
<p>Second Paragraph</p>

<script>
   let paragraphs = document.getElementsByTagName("p");
   console.log(paragraphs);
</script>
```

The method returns to a collection containing all matching elements.

Note

Name	Main Purpose
document.write()	Writes content to the document.
document.writeln()	Writes content with a newline.
document.getElementById()	Finds an element using its id.
document.getElementsByClassName()	Finds elements using a class name.
document.getElementsByTagName()	Finds elements using a tag name.

Properties

Properties are used to read or change information associated with an HTML element.

1. innerHTML

The innerHTML property gets or changes the HTML content inside an element.

```
<p id="message">Hello</p>

<script>
   document.getElementById("message").innerHTML = "<b>Welcome</b>";
</script>
```

The content inside the paragraph becomes bold because HTML markup is interpreted.

Reading innerHTML

```
let content = document.getElementById("message").innerHTML;
console.log(content);
```

innerHTML can read the HTML markup contained inside an element.

2. innerText

The innerText property gets or changes the visible text inside an element.

```
<p id="message">Hello Students</p>

<script>
   document.getElementById("message").innerText = "Welcome Students";
</script>
```

The visible text changes to Welcome Students.

innerHTML vs innerText

Property	Purpose	Example
innerHTML	Gets or sets HTML content inside an element.	element.innerHTML = "Hello";
innerText	Gets or sets visible text inside an element.	element.innerText = "Hello";